

Time Series Transformations in R: Moving Averages, SAX, XSAX

Roadmap

Let's explore practical examples of preprocessing with the **Harbinger** package in R — from smoothing with moving averages to symbolic representations (SAX and XSAX).

Smoothing

Moving average smoothing with `mas()`

SAX

Symbolic representation with `trans_sax()`

XSAX

Extended alphabet representation with `trans_xsax()`

📄 Run the complete examples and explore the source code at: <https://github.com/cefet-rj-dal/harbinger/tree/master/examples>

Moving Average Smoothing

What is it? A smoothing technique that replaces each value in a time series with the average of its neighboring values within a window of size k (the **order** parameter). This reduces local noise while preserving the overall trend.

How it works: For each point t , the smoothed value is the mean of the k surrounding observations. Larger windows produce smoother signals but may blur sharp transitions.

When to use it: Before applying event detectors, to reduce high-frequency noise that could trigger false positives.

Raw Signal

Noisy time series input

Sliding Window

Average computed over k neighbors

Smoothed Output

Cleaner signal ready for detection

Set Up the Example: mas()

```
# Install Harbinger (if needed)
# install.packages("harbinger")

# Load packages
library(harbinger)

# Load change-point datasets
data(examples_changepoints)

# Select simple dataset
dataset <- examples_changepoints$simple
head(dataset, 3)

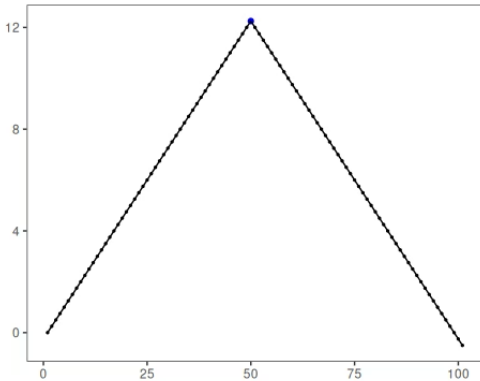
## serie event
## 1  0.00 FALSE
## 2  0.25 FALSE
## 3  0.50 FALSE
```

Step by Step

- Install the `harbinger` package if it is not yet available
- Load the package with `library(harbinger)`
- Use `data(examples_changepoints)` to access the built-in example datasets
- Select the `simple` dataset — it features a clear, well-defined level shift
- `head()` previews the first rows: the `serie` column holds the signal values and `event` holds the change-point markers

Apply `mas()` to the Series

```
# Plot the original time series
har_plot(
  harbinger(),
  dataset$serie,
  event = dataset$event
)
```



```
# Calculate smoothers with two window sizes
ma_5 <- mas(dataset$serie, order = 5)
ma_15 <- mas(dataset$serie, order = 15)
```

```
# Inspect the first smoothed values
head(ma_5)
## Time Series: Start=1, End=6, Freq=1
## [1] 0.50 0.75 1.00 1.25 1.50 1.75

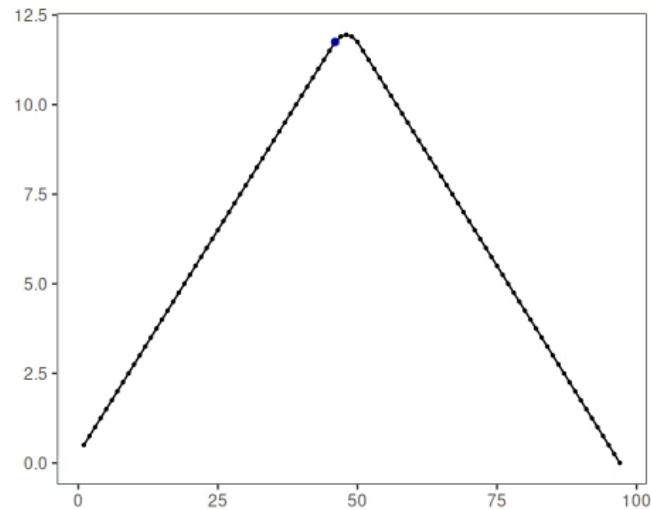
head(ma_15)
## Time Series: Start=1, End=6, Freq=1
## [1] 1.75 2.00 2.25 2.50 2.75 3.00
```

Step by step

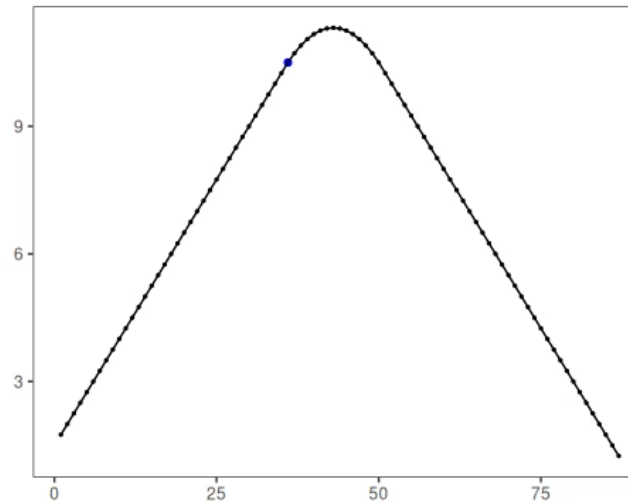
- `mas(serie, order = k)` computes the simple moving average with a window of size `k`
- `order = 5`: small window — preserves more local detail
- `order = 15`: large window — more aggressive smoothing, higher values
- Notice that the first values differ: larger windows shift the series further in time
- The return value is an R `ts` (time series) object

Visualizing the Smoothed Series

```
# Plot 5-point moving average
har_plot(
  harbinger(),
  as.numeric(ma_5),
  event = dataset$event[5:length(dataset$event)]
)
```



```
# Plot 15-point moving average
har_plot(
  harbinger(),
  as.numeric(ma_15),
  event =
    dataset$event[15:length(dataset$event)]
)
```



Step by step

- `as.numeric()` converts the `ts` object into a plain numeric vector
- The event vector is trimmed with `[k:length(...)]` to align with the smoothed series, which has `k-1` fewer points
- Compare the two plots: `order=5` keeps the transition clearly visible; `order=15` dilutes it further
- Smoothing is often used as a preprocessing step before applying change-point detectors

📌 **Key concept:** Smoothing before detection reduces false positives caused by local noise — but windows that are too large can mask genuine change points.

SAX – Symbolic Aggregate approXimation

What is it? SAX is a symbolic representation of time series that converts a continuous numeric signal into a sequence of discrete symbols (letters), enabling pattern matching, indexing, and dimensionality reduction.

How it works: The series is first z-normalized (zero mean, unit variance). It is then divided into equal-length segments, each replaced by its mean (PAA — Piecewise Aggregate Approximation). Finally, each PAA value is mapped to a symbol based on breakpoints derived from a Gaussian distribution, producing a string of **alpha** symbols.

Key parameter: **alpha** — the alphabet size. Larger alphabets capture more detail; smaller alphabets yield more compact representations.

When to use it: When you need to compare, cluster, or index time series by shape rather than exact values.

Z-Normalization

zero mean, unit variance

PAA Segmentation

divide into equal windows, compute means

Symbol Mapping

assign letters via Gaussian breakpoints → string output

- ❏ "Lin, J., Keogh, E., Wei, L., Lonardi, S. (2007). Experiencing SAX: a novel symbolic representation of time series. Data Mining and Knowledge Discovery, 15(2), 107–144. doi:10.1007/s10618-007-0064-z"

Setting Up the Example: trans_sax()

```
# Load required packages
library(daltoolbox)
library(harbinger)

# Load motif datasets
data(examples_motifs)

# Select the simple dataset
dataset <- examples_motifs$simple
head(dataset)

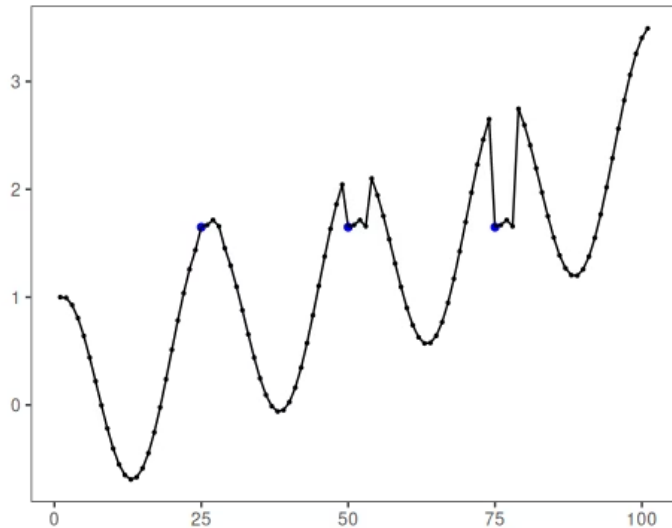
## serie   event
## 1  1.0000000 FALSE
## 2  0.9939124 FALSE
## 3  0.9275826 FALSE
## 4  0.8066889 FALSE
## 5  0.6403023 FALSE
```

What to note

- SAX requires `daltoolbox` — load it before `harbinger`
- The `simple` dataset is an oscillatory series with values between -1 and 1
- The `event` column marks positions of interest for later validation

Visualize and Configure trans_sax()

```
# Plot the original series
har_plot(
  harbinger(),
  dataset$serie,
  event = dataset$event
)
```



```
# Configure the SAX transformer
model <- trans_sax(alpha = 8)

# Fit and transform the series
model <- fit(model, dataset$serie)
sax_series <- transform(model, dataset$serie)

# Inspect the first symbols
head(sax_series, 20)
## [1] "D" "D" "D" "C" "C" "B" "B" "B"
##  "A" "A" "A" "A" "A" "A" "A" "A"
##  "A" "A" "B" "B"
```

Step-by-step

- `trans_sax(alpha = 8)` creates a transformer with an 8-letter alphabet (A–H)
- `fit(model, serie)` estimates the discretization breakpoints via z-normalization
- `transform(model, serie)` applies the transformation and returns symbols
- High values (~ 1.0) map to "D"; low values (~ -0.7) map to "A"
- The result is a character sequence — ideal for motif and discord analysis

❏ **Key concept:** The `alpha` parameter defines the alphabet size: more letters means a finer partition of the numeric values.

SAX vs XSAX: Symbol Mapping Comparison

SAX (alpha = 8) – Letters A–D

```
# Build comparison table for SAX
comparison <- data.frame(
  idx = seq_along(dataset$serie),
  value = dataset$serie,
  sax = sax_series
)
head(comparison, 5)

## idx    value sax
## 1  1  1.00000000 D
## 2  2  0.99391242 D
## 3  3  0.92758256 D
## 4  4  0.80668887 C
## 5  5  0.64030231 C
```

4 distinct levels (A → D) for the full range ~ -0.77 to ~ 1.0

XSAX (alpha = 16) – Digits 0–6

```
# Build comparison table for XSAX
model_x <- trans_xsax(alpha = 16)
model_x <- fit(model_x, dataset$serie)
xsax_series <- transform(model_x, dataset$serie)

comparison_x <- data.frame(
  idx = seq_along(dataset$serie),
  value = dataset$serie,
  xsax = xsax_series
)
head(comparison_x, 5)

## idx    value xsax
## 1  1  1.00000000  6
## 2  2  0.99391242  6
## 3  3  0.92758256  6
## 4  4  0.80668887  5
## 5  5  0.64030231  4
```

7 distinct levels (0 → 6) for the same range ~ -0.77 to ~ 1.0

Step by Step

- Both tables use the **same dataset** and identical numeric values — only the encoding differs
- SAX** (alpha = 8): descent from ~ 1.0 to ~ -0.7 yields only D → C → B → A — just 4 levels
- XSAX** (alpha = 16): same descent yields 6 → 5 → 4 → 3 → 2 → 1 → 0 — 7 distinct levels
- XSAX captures **finer amplitude differences** — valuable when motifs differ subtly in magnitude
- At index 19–20, SAX gives B both times; XSAX distinguishes 2 vs 3, preserving the upward trend

 **Reference:** Lin et al. (2007). A symbolic representation of time series. *Data Mining and Knowledge Discovery*, 15, 107–144.

XSAX – Extended Symbolic Representation

What is it? XSAX is an extension of SAX that uses a larger alphanumeric alphabet (digits 0–9 and letters A–F, up to 16 symbols), enabling finer-grained symbolic partitioning of the signal.

How it works: Like SAX, XSAX z-normalizes the series and applies PAA segmentation. The key difference is the alphabet: instead of a small set of letters (e.g., a–d), XSAX maps values to a 16-symbol alphanumeric set, capturing more intermediate levels of the signal.

Key parameter: `alpha = 16` — uses the full extended alphabet. Can be reduced for coarser representations.

SAX vs. XSAX: SAX is ideal for coarse shape matching; XSAX is better when finer amplitude distinctions matter (e.g., detecting subtle motifs or gradual drifts).

When to use it: When SAX's alphabet is too coarse and you need higher symbolic resolution without returning to raw numeric values.

SAX

small alphabet (e.g., a–d)

coarse partitioning

compact representation

XSAX

large alphabet (0–9, A–F)

fine partitioning

higher symbolic resolution

Configure and Apply `trans_xsax()`

```
# Reuse the same motifs dataset
# dataset <- examples_motifs$simple

# Configure the XSAX transformer
model <- trans_xsax(alpha = 16)

# Fit and transform
model <- fit(model, dataset$serie)
xsax_series <- transform(model,
dataset$serie)
```

```
# Inspect the first symbols
head(xsax_series, 20)
## [1] "6" "6" "6" "5" "4" "3" "2" "2"
##  "1" "0" "0" "0" "0" "0" "0" "0"
##  "1" "1" "2" "3"
```

Step by Step

- XSAX follows the same logic as SAX: it normalizes and discretizes the series
- `trans_xsax(alpha = 16)` uses a larger alphanumeric alphabet (0–9, A–F)
- The workflow is identical: `fit()` → `transform()`
- High values (~1.0) map to "6"; low values (~-0.7) map to "0"
- Compared to SAX, XSAX distinguishes more intermediate levels (e.g., 3, 4, 5 vs. just B, C)



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